installing libraries

In [1]:
```
pip install numpy
```

```
Requirement already satisfied: numpy in c:\users\lenovo\anaconda3\lib\site-packag
es (2.1.3)Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\lenovo\anaconda3\lib\site-packa
ges (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\lenovo\anaconda
3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site
-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\si
te-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
pip install snowflake-connector-python
```

```
Requirement already satisfied: snowflake-connector-python in c:\users\lenovo\anac
onda3\lib\site-packages (4.3.0)
Requirement already satisfied: asn1crypto<2.0.0,>0.24.0 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python) (1.5.1)
Requirement already satisfied: cryptography>=44.0.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python) (44.0.1)
Requirement already satisfied: pyOpenSSL<26.0.0,>=24.0.0 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python) (25.0.0)
Requirement already satisfied: pyjwt<3.0.0,>=2.10.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python) (2.10.1)
Requirement already satisfied: pytz in c:\users\lenovo\anaconda3\lib\site-package
s (from snowflake-connector-python) (2024.1)
Requirement already satisfied: requests<3.0.0,>=2.32.4 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python) (2.32.5)
Requirement already satisfied: packaging in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from snowflake-connector-python) (24.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python) (3.3.2)
Requirement already satisfied: idna<4,>=3.7 in c:\users\lenovo\anaconda3\lib\site
-packages (from snowflake-connector-python) (3.7)
Requirement already satisfied: certifi>=2024.7.4 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python) (2025.8.3)
Requirement already satisfied: typing_extensions<5,>=4.3 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python) (4.15.0)
Requirement already satisfied: filelock<4,>=3.5 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python) (3.17.0)
Requirement already satisfied: sortedcontainers>=2.4.0 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python) (2.4.0)
Requirement already satisfied: platformdirs<5.0.0,>=2.6.0 in c:\users\lenovo\anac
onda3\lib\site-packages (from snowflake-connector-python) (4.3.7)
Requirement already satisfied: tomlkit in c:\users\lenovo\anaconda3\lib\site-pack
ages (from snowflake-connector-python) (0.13.2)
Requirement already satisfied: boto3>=1.24 in c:\users\lenovo\anaconda3\lib\site-
packages (from snowflake-connector-python) (1.42.59)
Requirement already satisfied: botocore>=1.24 in c:\users\lenovo\anaconda3\lib\si
te-packages (from snowflake-connector-python) (1.42.59)
Requirement already satisfied: cffi>=1.12 in c:\users\lenovo\anaconda3\lib\site-p
ackages (from cryptography>=44.0.1->snowflake-connector-python) (1.17.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\li
b\site-packages (from requests<3.0.0,>=2.32.4->snowflake-connector-python) (2.3.
0)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in c:\users\lenovo\anaconda
3\lib\site-packages (from boto3>=1.24->snowflake-connector-python) (1.0.1)
Requirement already satisfied: s3transfer<0.17.0,>=0.16.0 in c:\users\lenovo\anac
onda3\lib\site-packages (from boto3>=1.24->snowflake-connector-python) (0.16.0)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in c:\users\lenovo\ana
conda3\lib\site-packages (from botocore>=1.24->snowflake-connector-python) (2.9.
0.post0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil<3.0.0,>=2.1->botocore>=1.24->snowflake-connector-pyth
on) (1.17.0)
Requirement already satisfied: pycparser in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from cffi>=1.12->cryptography>=44.0.1->snowflake-connector-python) (2.21)
Note: you may need to restart the kernel to use updated packages.
```

In [4]:
```
pip install jupyter_scheduler
```

```
Requirement already satisfied: jupyter_scheduler in c:\users\lenovo\anaconda3\lib
\site-packages (2.11.0)
Requirement already satisfied: croniter~=1.4 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from jupyter_scheduler) (1.4.1)
Requirement already satisfied: fsspec!=2025.3.1,<=2025.3.2,>=2023.6.0 in c:\users
\lenovo\anaconda3\lib\site-packages (from jupyter_scheduler) (2025.3.2)
Requirement already satisfied: jupyter-server<3,>=1.6 in c:\users\lenovo\anaconda
3\lib\site-packages (from jupyter_scheduler) (2.15.0)
Requirement already satisfied: nbconvert~=7.0 in c:\users\lenovo\anaconda3\lib\si
te-packages (from jupyter_scheduler) (7.16.6)
Requirement already satisfied: psutil~=5.9 in c:\users\lenovo\anaconda3\lib\site-
packages (from jupyter_scheduler) (5.9.0)
Requirement already satisfied: pydantic<3,>=1.10 in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter_scheduler) (2.10.3)
Requirement already satisfied: pytz<=2024.2,>=2023.3 in c:\users\lenovo\anaconda3
\lib\site-packages (from jupyter_scheduler) (2024.1)
Requirement already satisfied: sqlalchemy<3,>=2.0 in c:\users\lenovo\anaconda3\li
b\site-packages (from jupyter_scheduler) (2.0.39)
Requirement already satisfied: traitlets~=5.0 in c:\users\lenovo\anaconda3\lib\si
te-packages (from jupyter_scheduler) (5.14.3)
Requirement already satisfied: python-dateutil in c:\users\lenovo\anaconda3\lib\s
ite-packages (from croniter~=1.4->jupyter_scheduler) (2.9.0.post0)
Requirement already satisfied: anyio>=3.1.0 in c:\users\lenovo\anaconda3\lib\site
-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (4.7.0)
Requirement already satisfied: argon2-cffi>=21.1 in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (21.3.0)
Requirement already satisfied: jinja2>=3.0.3 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (3.1.6)
Requirement already satisfied: jupyter-client>=7.4.4 in c:\users\lenovo\anaconda3
\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (8.6.3)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in c:\users\lenovo\anac
onda3\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (5.7.2)
Requirement already satisfied: jupyter-events>=0.11.0 in c:\users\lenovo\anaconda
3\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (0.12.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in c:\users\lenovo
\anaconda3\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (0.
5.3)
Requirement already satisfied: nbformat>=5.3.0 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (5.10.4)
Requirement already satisfied: overrides>=5.0 in c:\users\lenovo\anaconda3\lib\si
te-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (7.4.0)
Requirement already satisfied: packaging>=22.0 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (24.2)
Requirement already satisfied: prometheus-client>=0.9 in c:\users\lenovo\anaconda
3\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (0.21.1)
Requirement already satisfied: pywinpty>=2.0.1 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (2.0.15)
Requirement already satisfied: pyzmq>=24 in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from jupyter-server<3,>=1.6->jupyter_scheduler) (26.2.0)
Requirement already satisfied: send2trash>=1.8.2 in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (1.8.2)
Requirement already satisfied: terminado>=0.8.3 in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (0.17.1)
Requirement already satisfied: tornado>=6.2.0 in c:\users\lenovo\anaconda3\lib\si
te-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (6.5.1)
Requirement already satisfied: websocket-client>=1.7 in c:\users\lenovo\anaconda3
\lib\site-packages (from jupyter-server<3,>=1.6->jupyter_scheduler) (1.8.0)
Requirement already satisfied: beautifulsoup4 in c:\users\lenovo\anaconda3\lib\si
te-packages (from nbconvert~=7.0->jupyter_scheduler) (4.12.3)
Requirement already satisfied: bleach!=5.0.0 in c:\users\lenovo\anaconda3\lib\sit
```

```
e-packages (from bleach[css]!=5.0.0->nbconvert~=7.0->jupyter_scheduler) (6.2.0)
Requirement already satisfied: defusedxml in c:\users\lenovo\anaconda3\lib\site-p
ackages (from nbconvert~=7.0->jupyter_scheduler) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in c:\users\lenovo\anaconda3\l
ib\site-packages (from nbconvert~=7.0->jupyter_scheduler) (0.3.0)
Requirement already satisfied: markupsafe>=2.0 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from nbconvert~=7.0->jupyter_scheduler) (3.0.2)
Requirement already satisfied: mistune<4,>=2.0.3 in c:\users\lenovo\anaconda3\lib
\site-packages (from nbconvert~=7.0->jupyter_scheduler) (3.1.2)
Requirement already satisfied: nbclient>=0.5.0 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from nbconvert~=7.0->jupyter_scheduler) (0.10.2)
Requirement already satisfied: pandocfilters>=1.4.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from nbconvert~=7.0->jupyter_scheduler) (1.5.0)
Requirement already satisfied: pygments>=2.4.1 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from nbconvert~=7.0->jupyter_scheduler) (2.19.1)
Requirement already satisfied: annotated-types>=0.6.0 in c:\users\lenovo\anaconda
3\lib\site-packages (from pydantic<3,>=1.10->jupyter_scheduler) (0.6.0)
Requirement already satisfied: pydantic-core==2.27.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from pydantic<3,>=1.10->jupyter_scheduler) (2.27.1)
Requirement already satisfied: typing-extensions>=4.12.2 in c:\users\lenovo\anaco
nda3\lib\site-packages (from pydantic<3,>=1.10->jupyter_scheduler) (4.15.0)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\lenovo\anaconda3\lib
\site-packages (from sqlalchemy<3,>=2.0->jupyter_scheduler) (3.1.1)
Requirement already satisfied: idna>=2.8 in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from anyio>=3.1.0->jupyter-server<3,>=1.6->jupyter_scheduler) (3.7)
Requirement already satisfied: sniffio>=1.1 in c:\users\lenovo\anaconda3\lib\site
-packages (from anyio>=3.1.0->jupyter-server<3,>=1.6->jupyter_scheduler) (1.3.0)
Requirement already satisfied: argon2-cffi-bindings in c:\users\lenovo\anaconda3
\lib\site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=1.6->jupyter_sched
uler) (21.2.0)
Requirement already satisfied: webencodings in c:\users\lenovo\anaconda3\lib\site
-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert~=7.0->jupyter_schedu
ler) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in c:\users\lenovo\anaconda3
\lib\site-packages (from bleach[css]!=5.0.0->nbconvert~=7.0->jupyter_scheduler)
(1.4.0)
Requirement already satisfied: platformdirs>=2.5 in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=1.6->jupyter_
scheduler) (4.3.7)
Requirement already satisfied: pywin32>=300 in c:\users\lenovo\anaconda3\lib\site
-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=1.6->jupyter_sched
uler) (308)
Requirement already satisfied: jsonschema>=4.18.0 in c:\users\lenovo\anaconda3\li
b\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->
jupyter-server<3,>=1.6->jupyter_scheduler) (4.23.0)
Requirement already satisfied: python-json-logger>=2.0.4 in c:\users\lenovo\anaco
nda3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupy
ter_scheduler) (3.2.1)
Requirement already satisfied: pyyaml>=5.3 in c:\users\lenovo\anaconda3\lib\site-
packages (from jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler)
(6.0.3)
Requirement already satisfied: referencing in c:\users\lenovo\anaconda3\lib\site-
packages (from jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler)
(0.30.2)
Requirement already satisfied: rfc3339-validator in c:\users\lenovo\anaconda3\lib
\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupyter_sche
duler) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in c:\users\lenovo\anacon
da3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupyt
er_scheduler) (0.1.1)
```

```
Requirement already satisfied: attrs>=22.2.0 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from jsonschema>=4.18.0->jsonschema[format-nongpl]>=4.18.0->jupyter-e
vents>=0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler) (24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\users\l
enovo\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jsonschema[format-non
gpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler)
(2023.7.1)
Requirement already satisfied: rpds-py>=0.7.1 in c:\users\lenovo\anaconda3\lib\si
te-packages (from jsonschema>=4.18.0->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler) (0.22.3)
Requirement already satisfied: fqdn in c:\users\lenovo\anaconda3\lib\site-package
s (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server
<3,>=1.6->jupyter_scheduler) (1.5.1)
Requirement already satisfied: isoduration in c:\users\lenovo\anaconda3\lib\site-
packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter
-server<3,>=1.6->jupyter_scheduler) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in c:\users\lenovo\anaconda3\lib
\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->j
upyter-server<3,>=1.6->jupyter_scheduler) (2.1)
Requirement already satisfied: uri-template in c:\users\lenovo\anaconda3\lib\site
-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyte
r-server<3,>=1.6->jupyter_scheduler) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in c:\users\lenovo\anaconda3\lib
\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->j
upyter-server<3,>=1.6->jupyter_scheduler) (25.10.0)
Requirement already satisfied: fastjsonschema>=2.15 in c:\users\lenovo\anaconda3
\lib\site-packages (from nbformat>=5.3.0->jupyter-server<3,>=1.6->jupyter_schedul
er) (2.20.0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil->croniter~=1.4->jupyter_scheduler) (1.17.0)
Requirement already satisfied: cffi>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-
packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=1.6->j
upyter_scheduler) (1.17.1)
Requirement already satisfied: pycparser in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server
<3,>=1.6->jupyter_scheduler) (2.21)
Requirement already satisfied: soupsieve>1.2 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from beautifulsoup4->nbconvert~=7.0->jupyter_scheduler) (2.5)
Requirement already satisfied: arrow>=0.15.0 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=
0.11.0->jupyter-server<3,>=1.6->jupyter_scheduler) (1.3.0)
Note: you may need to restart the kernel to use updated packages.
```

In [5]:
```
pip install snowflake-sqlalchemy
```

```
Requirement already satisfied: snowflake-sqlalchemy in c:\users\lenovo\anaconda3
\lib\site-packages (1.8.2)
Requirement already satisfied: snowflake-connector-python<5.0.0 in c:\users\lenov
o\anaconda3\lib\site-packages (from snowflake-sqlalchemy) (4.3.0)
Requirement already satisfied: sqlalchemy>=1.4.19 in c:\users\lenovo\anaconda3\li
b\site-packages (from snowflake-sqlalchemy) (2.0.39)
Requirement already satisfied: asn1crypto<2.0.0,>0.24.0 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchem
y) (1.5.1)
Requirement already satisfied: cryptography>=44.0.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy)
(44.0.1)
Requirement already satisfied: pyOpenSSL<26.0.0,>=24.0.0 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalche
my) (25.0.0)
Requirement already satisfied: pyjwt<3.0.0,>=2.10.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy)
(2.10.1)
Requirement already satisfied: pytz in c:\users\lenovo\anaconda3\lib\site-package
s (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (2024.1)
Requirement already satisfied: requests<3.0.0,>=2.32.4 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchem
y) (2.32.5)
Requirement already satisfied: packaging in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (24.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchem
y) (3.3.2)
Requirement already satisfied: idna<4,>=3.7 in c:\users\lenovo\anaconda3\lib\site
-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (3.7)
Requirement already satisfied: certifi>=2024.7.4 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (202
5.8.3)
Requirement already satisfied: typing_extensions<5,>=4.3 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalche
my) (4.15.0)
Requirement already satisfied: filelock<4,>=3.5 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (3.1
7.0)
Requirement already satisfied: sortedcontainers>=2.4.0 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchem
y) (2.4.0)
Requirement already satisfied: platformdirs<5.0.0,>=2.6.0 in c:\users\lenovo\anac
onda3\lib\site-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalch
emy) (4.3.7)
Requirement already satisfied: tomlkit in c:\users\lenovo\anaconda3\lib\site-pack
ages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (0.13.2)
Requirement already satisfied: boto3>=1.24 in c:\users\lenovo\anaconda3\lib\site-
packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (1.42.59)
Requirement already satisfied: botocore>=1.24 in c:\users\lenovo\anaconda3\lib\si
te-packages (from snowflake-connector-python<5.0.0->snowflake-sqlalchemy) (1.42.5
9)
Requirement already satisfied: cffi>=1.12 in c:\users\lenovo\anaconda3\lib\site-p
ackages (from cryptography>=44.0.1->snowflake-connector-python<5.0.0->snowflake-s
qlalchemy) (1.17.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\li
b\site-packages (from requests<3.0.0,>=2.32.4->snowflake-connector-python<5.0.0->
snowflake-sqlalchemy) (2.3.0)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in c:\users\lenovo\anaconda
3\lib\site-packages (from boto3>=1.24->snowflake-connector-python<5.0.0->snowflak
```

```
e-sqlalchemy) (1.0.1)
Requirement already satisfied: s3transfer<0.17.0,>=0.16.0 in c:\users\lenovo\anac
onda3\lib\site-packages (from boto3>=1.24->snowflake-connector-python<5.0.0->snow
flake-sqlalchemy) (0.16.0)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in c:\users\lenovo\ana
conda3\lib\site-packages (from botocore>=1.24->snowflake-connector-python<5.0.0->
snowflake-sqlalchemy) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil<3.0.0,>=2.1->botocore>=1.24->snowflake-connector-pyth
on<5.0.0->snowflake-sqlalchemy) (1.17.0)
Requirement already satisfied: pycparser in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from cffi>=1.12->cryptography>=44.0.1->snowflake-connector-python<5.0.0->
snowflake-sqlalchemy) (2.21)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\lenovo\anaconda3\lib
\site-packages (from sqlalchemy>=1.4.19->snowflake-sqlalchemy) (3.1.1)
Note: you may need to restart the kernel to use updated packages.
```

In [6]:
```
pip install "snowflake-connector-python[pandas]"
```

```
Requirement already satisfied: snowflake-connector-python[pandas] in c:\users\len
ovo\anaconda3\lib\site-packages (4.3.0)Note: you may need to restart the kernel t
o use updated packages.

Requirement already satisfied: asn1crypto<2.0.0,>0.24.0 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python[pandas]) (1.5.1)
Requirement already satisfied: cryptography>=44.0.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python[pandas]) (44.0.1)
Requirement already satisfied: pyOpenSSL<26.0.0,>=24.0.0 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python[pandas]) (25.0.0)
Requirement already satisfied: pyjwt<3.0.0,>=2.10.1 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python[pandas]) (2.10.1)
Requirement already satisfied: pytz in c:\users\lenovo\anaconda3\lib\site-package
s (from snowflake-connector-python[pandas]) (2024.1)
Requirement already satisfied: requests<3.0.0,>=2.32.4 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python[pandas]) (2.32.5)
Requirement already satisfied: packaging in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from snowflake-connector-python[pandas]) (24.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\lenovo\anacon
da3\lib\site-packages (from snowflake-connector-python[pandas]) (3.3.2)
Requirement already satisfied: idna<4,>=3.7 in c:\users\lenovo\anaconda3\lib\site
-packages (from snowflake-connector-python[pandas]) (3.7)
Requirement already satisfied: certifi>=2024.7.4 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python[pandas]) (2025.8.3)
Requirement already satisfied: typing_extensions<5,>=4.3 in c:\users\lenovo\anaco
nda3\lib\site-packages (from snowflake-connector-python[pandas]) (4.15.0)
Requirement already satisfied: filelock<4,>=3.5 in c:\users\lenovo\anaconda3\lib
\site-packages (from snowflake-connector-python[pandas]) (3.17.0)
Requirement already satisfied: sortedcontainers>=2.4.0 in c:\users\lenovo\anacond
a3\lib\site-packages (from snowflake-connector-python[pandas]) (2.4.0)
Requirement already satisfied: platformdirs<5.0.0,>=2.6.0 in c:\users\lenovo\anac
onda3\lib\site-packages (from snowflake-connector-python[pandas]) (4.3.7)
Requirement already satisfied: tomlkit in c:\users\lenovo\anaconda3\lib\site-pack
ages (from snowflake-connector-python[pandas]) (0.13.2)
Requirement already satisfied: boto3>=1.24 in c:\users\lenovo\anaconda3\lib\site-
packages (from snowflake-connector-python[pandas]) (1.42.59)
Requirement already satisfied: botocore>=1.24 in c:\users\lenovo\anaconda3\lib\si
te-packages (from snowflake-connector-python[pandas]) (1.42.59)
Requirement already satisfied: pandas<3.0.0,>=2.1.2 in c:\users\lenovo\anaconda3
\lib\site-packages (from snowflake-connector-python[pandas]) (2.2.3)
Requirement already satisfied: pyarrow>=14.0.1 in c:\users\lenovo\anaconda3\lib\s
ite-packages (from snowflake-connector-python[pandas]) (19.0.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\lenovo\anaconda3\lib\sit
e-packages (from pandas<3.0.0,>=2.1.2->snowflake-connector-python[pandas]) (2.1.
3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\lenovo\anaconda
3\lib\site-packages (from pandas<3.0.0,>=2.1.2->snowflake-connector-python[panda
s]) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\si
te-packages (from pandas<3.0.0,>=2.1.2->snowflake-connector-python[pandas]) (202
5.2)
Requirement already satisfied: cffi>=1.12 in c:\users\lenovo\anaconda3\lib\site-p
ackages (from cryptography>=44.0.1->snowflake-connector-python[pandas]) (1.17.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\li
b\site-packages (from requests<3.0.0,>=2.32.4->snowflake-connector-python[panda
s]) (2.3.0)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in c:\users\lenovo\anaconda
3\lib\site-packages (from boto3>=1.24->snowflake-connector-python[pandas]) (1.0.
1)
Requirement already satisfied: s3transfer<0.17.0,>=0.16.0 in c:\users\lenovo\anac
```

```
onda3\lib\site-packages (from boto3>=1.24->snowflake-connector-python[pandas])
(0.16.0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-pac
kages (from python-dateutil>=2.8.2->pandas<3.0.0,>=2.1.2->snowflake-connector-pyt
hon[pandas]) (1.17.0)
Requirement already satisfied: pycparser in c:\users\lenovo\anaconda3\lib\site-pa
ckages (from cffi>=1.12->cryptography>=44.0.1->snowflake-connector-python[panda
s]) (2.21)
```

Connecting to Snowflake

In [2]:
```
import numpy as np
import pandas as pd
import pandas_profiling
import matplotlib.pyplot as plt
import getpass
import seaborn as sns
%matplotlib inline
import snowflake.connector
```

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_8752\3477197541.py:3: DeprecationWar
ning: `import pandas_profiling` is going to be deprecated by April 1st. Please us
e `import ydata_profiling` instead.
  import pandas_profiling
```

In [160…
```
conn = snowflake.connector.connect(
        user = 'Moni',
        password = getpass.getpass('Your Snowflake Password is'),
        account = 'CNXLYPS-NB32912',
        database = 'RETAILS',
        schema = 'PUBLIC',
        warehouse = 'COMPUTE_WH',
)
```

In [161…
```
cur = conn.cursor()
```

In [162…
```
cur
```

Out[162…
```
<snowflake.connector.cursor.SnowflakeCursor at 0x22eb91a87d0>
```

In [163…
```
select_demographic_RAW = 'SELECT * FROM demographic_RAW'
select_CAMPAIGN_DESC_RAW = 'SELECT * FROM CAMPAIGN_DESC_RAW'
select_CAMPAIGN_RAW = 'SELECT * FROM CAMPAIGN_RAW'
select_PRODUCT_RAW = 'SELECT * FROM PRODUCT_RAW'
select_COUPON_RAW = 'SELECT * FROM COUPON_RAW'
select_COUPON_REDEMPT_RAW = 'SELECT * FROM COUPON_REDEMPT_RAW'
select_TRANSACTION_RAW = 'SELECT * FROM TRANSACTION_RAW'
```

In [164…
```
cur.execute(select_demographic_RAW)
demographic_RAW = cur.fetch_pandas_all()
```

In [165…
```
cur.execute(select_CAMPAIGN_DESC_RAW)
CAMPAIGN_DESC_RAW = cur.fetch_pandas_all()
```

In [166…
```
cur.execute(select_CAMPAIGN_RAW)
CAMPAIGN_RAW = cur.fetch_pandas_all()
```

In [167…
```
cur.execute(select_PRODUCT_RAW)
PRODUCT_RAW = cur.fetch_pandas_all()
```

In [168…
```
cur.execute(select_COUPON_RAW)
COUPON_RAW = cur.fetch_pandas_all()
```

In [169…
```
cur.execute(select_COUPON_REDEMPT_RAW)
COUPON_REDEMPT_RAW = cur.fetch_pandas_all()
```

In [170…
```
cur.execute(select_TRANSACTION_RAW)
TRANSACTION_RAW = cur.fetch_pandas_all()
```

In [171…
```
cur.close()
conn.close()
```

In [172…
```
demographic_RAW.head(5)
```

Out[172…

	AGE_DESC	MARITAL_STATUS_CODE	INCOME_DESC	HOMEOWNER_DESC	HH_COMP_I
0	65+	A	35-49K	Homeowner	2 Adults No
1	45-54	A	50-74K	Homeowner	2 Adults No
2	25-34	U	25-34K	Unknown	2 Adults
3	25-34	U	75-99K	Homeowner	2 Adults
4	45-54	B	50-74K	Homeowner	Single Fe

In [173…
```
CAMPAIGN_DESC_RAW.head(5)
```

Out[173…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY
0	TypeB	24	659	719
1	TypeC	15	547	708
2	TypeB	25	659	691
3	TypeC	20	615	685
4	TypeB	23	646	684

In [174…
```
CAMPAIGN_RAW.head(5)
```

Out[174…

	DESCRIPTION	HOUSEHOLD_KEY	CAMPAIGN
0	TypeA	17	26
1	TypeA	27	26
2	TypeA	212	26
3	TypeA	208	26
4	TypeA	192	26

In [175…
```
PRODUCT_RAW.head(5)
```

Out[175…

	PRODUCT_ID	MANUFACTURER	DEPARTMENT	BRAND	COMMODITY_DESC	SUB_CC
0	25671	2	GROCERY	National	FRZN ICE	ICE -
1	26081	2	MISC. TRANS.	National	NO COMMODITY DESCRIPTION	NO
2	26093	69	PASTRY	Private	BREAD	BREAD
3	26190	69	GROCERY	Private	FRUIT - SHELF STABLE	
4	26355	69	GROCERY	Private	COOKIES/CONES	SF

In [176…
```
COUPON_RAW.head(5)
```

Out[176…

	COUPON_UPC	PRODUCT_ID	CAMPAIGN
0	10000089061	27160	4
1	10000089064	27754	9
2	10000089073	28897	12
3	51800009050	28919	28
4	52100000076	28929	25

In [177…
```
COUPON_REDEMPT_RAW.head(5)
```

Out[177…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN
0	1	421	10000085364	8
1	1	421	51700010076	8
2	1	427	54200000033	8
3	1	597	10000085476	18
4	1	597	54200029176	18

In [178…
```
TRANSACTION_RAW.head(5)
```

Out[178…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE	STOR
0	2375	26984851472	1	1004906	1	1.39	
1	2375	26984851472	1	1033142	1	0.82	
2	2375	26984851472	1	1036325	1	0.99	
3	2375	26984851472	1	1082185	1	1.21	
4	2375	26984851472	1	8160430	1	1.50	

In [179…
```
demographic_RAW.shape
```
Out[179…
```
(2500, 8)
```

In [180…
```
CAMPAIGN_DESC_RAW.shape
```
Out[180…
```
(30, 4)
```

In [181…
```
CAMPAIGN_RAW.shape
```
Out[181…
```
(7208, 3)
```

In [182…
```
PRODUCT_RAW.shape
```
Out[182…
```
(92353, 7)
```

In [183…
```
COUPON_RAW.shape
```
Out[183…
```
(124548, 3)
```

In [184…
```
COUPON_REDEMPT_RAW.shape
```
Out[184…
```
(2318, 4)
```

In [185…
```
TRANSACTION_RAW.shape
```
Out[185…
```
(450416, 12)
```

In [186…
```
demographic_RAW.dtypes
```
Out[186…
```
AGE_DESC               object
MARITAL_STATUS_CODE    object
INCOME_DESC            object
HOMEOWNER_DESC         object
HH_COMP_DESC           object
HOUSEHOLD_SIZE_DESC    object
KID_CATEGORY_DESC      object
HOUSEHOLD_KEY           int16
dtype: object
```

In [187…
```
demographic_RAW.describe()
```

Out[187…

	HOUSEHOLD_KEY
count	2500.00000
mean	1250.50000
std	721.83216
min	1.00000
25%	625.75000
50%	1250.50000
75%	1875.25000
max	2500.00000

In [188…

```
demographic_RAW.isnull().sum()
```

Out[188…

```
AGE_DESC               0
MARITAL_STATUS_CODE    0
INCOME_DESC            0
HOMEOWNER_DESC         0
HH_COMP_DESC           0
HOUSEHOLD_SIZE_DESC    0
KID_CATEGORY_DESC      0
HOUSEHOLD_KEY          0
dtype: int64
```

In [189…

```
PRODUCT_RAW.isnull().sum()
```

Out[189…

```
PRODUCT_ID              0
MANUFACTURER            0
DEPARTMENT              0
BRAND                   0
COMMODITY_DESC          0
SUB_COMMODITY_DESC      0
CURR_SIZE_OF_PRODUCT    0
dtype: int64
```

In [190…

```
PRODUCT_RAW.dtypes
```

Out[190…

```
PRODUCT_ID               int32
MANUFACTURER             int16
DEPARTMENT              object
BRAND                   object
COMMODITY_DESC          object
SUB_COMMODITY_DESC      object
CURR_SIZE_OF_PRODUCT    object
dtype: object
```

In [191…

```
TRANSACTION_RAW.isnull().sum()
```

Out[191…

```
HOUSEHOLD_KEY        0
BASKET_ID            0
DAY                  0
PRODUCT_ID           0
QUANTITY             0
SALES_VALUE          0
STORE_ID             0
RETAIL_DISC          0
TRANS_TIME           0
WEEK_NO              0
COUPON_DISC          0
COUPON_MATCH_DISC    0
dtype: int64
```

In [192…

```
TRANSACTION_RAW.dtypes
```

Out[192…

```
HOUSEHOLD_KEY          int16
BASKET_ID              int64
DAY                    int16
PRODUCT_ID             int32
QUANTITY               int32
SALES_VALUE          float64
STORE_ID               int16
RETAIL_DISC          float64
TRANS_TIME             int16
WEEK_NO                 int8
COUPON_DISC             int8
COUPON_MATCH_DISC       int8
dtype: object
```

In [193…

```
TRANSACTION_RAW.describe()
```

Out[193…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	S
count	450416.00000	4.504160e+05	450416.000000	4.504160e+05	450416.000000	45
mean	1259.55822	2.840143e+10	112.470718	2.209464e+06	86.522561	
std	728.10869	5.792110e+08	42.064258	2.726512e+06	1058.788785	
min	1.00000	2.698485e+10	1.000000	2.567100e+04	0.000000	
25%	641.00000	2.798531e+10	84.000000	9.118780e+05	1.000000	
50%	1234.00000	2.839168e+10	117.000000	1.016800e+06	1.000000	
75%	1914.00000	2.886773e+10	147.000000	1.115360e+06	1.000000	
max	2500.00000	2.933965e+10	176.000000	1.260019e+07	51912.000000	

In [194…

```
CAMPAIGN_DESC_RAW.isnull().sum()
```

Out[194…

```
DESCRIPTION    0
CAMPAIGN       0
START_DAY      0
END_DAY        0
dtype: int64
```

In [195…

```
CAMPAIGN_DESC_RAW.dtypes
```

Out[195…

```
DESCRIPTION    object
CAMPAIGN         int8
START_DAY       int16
END_DAY         int16
dtype: object
```

In [196…

```
CAMPAIGN_DESC_RAW.describe()
```

Out[196…

	CAMPAIGN	START_DAY	END_DAY
count	30.000000	30.000000	30.000000
mean	15.500000	463.866667	510.466667
std	8.803408	134.488490	137.730555
min	1.000000	224.000000	264.000000
25%	8.250000	360.000000	405.750000
50%	15.500000	470.000000	502.000000
75%	22.750000	584.000000	640.250000
max	30.000000	659.000000	719.000000

In [197…

```
COUPON_RAW.isnull().sum()
```

Out[197…

```
COUPON_UPC    0
PRODUCT_ID    0
CAMPAIGN      0
dtype: int64
```

In [198…

```
COUPON_RAW.dtypes
```

Out[198…

```
COUPON_UPC    int64
PRODUCT_ID    int32
CAMPAIGN       int8
dtype: object
```

In [199…

```
COUPON_REDEMPT_RAW.isnull().sum()
```

Out[199…

```
HOUSEHOLD_KEY    0
DAY              0
COUPON_UPC       0
CAMPAIGN         0
dtype: int64
```

In [200…

```
COUPON_REDEMPT_RAW.describe()
```

Out[200…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN
count	2318.000000	2318.000000	2.318000e+03	2318.000000
mean	1302.816652	528.217429	4.123049e+10	15.551337
std	783.002545	100.360985	1.986068e+10	5.716636
min	1.000000	225.000000	1.000009e+10	1.000000
25%	588.000000	458.250000	1.000009e+10	13.000000
50%	1396.500000	532.000000	5.234003e+10	14.000000
75%	2004.000000	605.000000	5.430002e+10	18.000000
max	2500.000000	704.000000	5.897850e+10	30.000000

Date Modification

In [201…

```
from datetime import datetime,timedelta
```

In [202…

```
start_date = pd.to_datetime('2019-01-01')
```

In [203…

```
start_date
```

Out[203…

```
Timestamp('2019-01-01 00:00:00')
```

Transaction Table

In [204…

```
TRANSACTION_RAW.head(5)
```

Out[204…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE	STOR
0	2375	26984851472	1	1004906	1	1.39	
1	2375	26984851472	1	1033142	1	0.82	
2	2375	26984851472	1	1036325	1	0.99	
3	2375	26984851472	1	1082185	1	1.21	
4	2375	26984851472	1	8160430	1	1.50	

In [205…

```
TRANSACTION_RAW['DATE'] =  start_date + pd.to_timedelta(TRANSACTION_RAW['DAY'],
```

In [206…

```
TRANSACTION_RAW['DATE']
```

Out[206…

```
0        2019-01-02
1        2019-01-02
2        2019-01-02
3        2019-01-02
4        2019-01-02
            ...
450411   2019-06-26
450412   2019-06-26
450413   2019-06-26
450414   2019-06-26
450415   2019-06-26
Name: DATE, Length: 450416, dtype: datetime64[ns]
```

In [207…

```
TRANSACTION_RAW['DATE'].head(20)
```

Out[207…

```
0    2019-01-02
1    2019-01-02
2    2019-01-02
3    2019-01-02
4    2019-01-02
5    2019-01-02
6    2019-01-02
7    2019-01-02
8    2019-01-02
9    2019-01-02
10   2019-01-02
11   2019-01-02
12   2019-01-02
13   2019-01-02
14   2019-01-02
15   2019-01-02
16   2019-01-02
17   2019-01-02
18   2019-01-02
19   2019-01-02
Name: DATE, dtype: datetime64[ns]
```

In [208…

```
TRANSACTION_RAW.head(5)
```

Out[208…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE	STOR
0	2375	26984851472	1	1004906	1	1.39	
1	2375	26984851472	1	1033142	1	0.82	
2	2375	26984851472	1	1036325	1	0.99	
3	2375	26984851472	1	1082185	1	1.21	
4	2375	26984851472	1	8160430	1	1.50	

Campaign Table

In [209…
```
CAMPAIGN_DESC_RAW.head(5)
```

Out[209…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY
0	TypeB	24	659	719
1	TypeC	15	547	708
2	TypeB	25	659	691
3	TypeC	20	615	685
4	TypeB	23	646	684

In [210…
```
CAMPAIGN_DESC_RAW['START_DAY'] = start_date + pd.to_timedelta(CAMPAIGN_DESC_RAW[
```

In [211…
```
CAMPAIGN_DESC_RAW['END_DAY'] = start_date + pd.to_timedelta(CAMPAIGN_DESC_RAW['E
```

In [212…
```
CAMPAIGN_DESC_RAW.head(5)
```

Out[212…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY
0	TypeB	24	2020-10-21	2020-12-20
1	TypeC	15	2020-07-01	2020-12-09
2	TypeB	25	2020-10-21	2020-11-22
3	TypeC	20	2020-09-07	2020-11-16
4	TypeB	23	2020-10-08	2020-11-15

In [213…
```
### Duration of campaign
CAMPAIGN_DESC_RAW['Campaign_Duration'] = CAMPAIGN_DESC_RAW['END_DAY'] - CAMPAIGN
```

In [214…
```
CAMPAIGN_DESC_RAW.head(5)
```

Out[214…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration
0	TypeB	24	2020-10-21	2020-12-20	60 days
1	TypeC	15	2020-07-01	2020-12-09	161 days
2	TypeB	25	2020-10-21	2020-11-22	32 days
3	TypeC	20	2020-09-07	2020-11-16	70 days
4	TypeB	23	2020-10-08	2020-11-15	38 days

Coupon Redempt Table

In [215…
```
COUPON_REDEMPT_RAW.head(5)
```

Out[215…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN
0	1	421	10000085364	8
1	1	421	51700010076	8
2	1	427	54200000033	8
3	1	597	10000085476	18
4	1	597	54200029176	18

In [216…

```
COUPON_REDEMPT_RAW['Date'] = start_date + pd.to_timedelta(COUPON_REDEMPT_RAW['DA
```

In [217…

```
COUPON_REDEMPT_RAW.head(5)
```

Out[217…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN	Date
0	1	421	10000085364	8	2020-02-26
1	1	421	51700010076	8	2020-02-26
2	1	427	54200000033	8	2020-03-03
3	1	597	10000085476	18	2020-08-20
4	1	597	54200029176	18	2020-08-20

In [218…

```
TRANSACTION_RAW['DATE'].max()              # latest transaction date
```

Out[218…

```
Timestamp('2019-06-26 00:00:00')
```

In [219…

```
CAMPAIGN_DESC_RAW['END_DAY'].max()
```

Out[219…

```
Timestamp('2020-12-20 00:00:00')
```

In [220…

```
COUPON_REDEMPT_RAW['Date'].max()          # last time coupon reedemed
```

Out[220…

```
Timestamp('2020-12-05 00:00:00')
```

Understand the data

In [221…

```
demographic_RAW.head(2)
```

Out[221…

	AGE_DESC	MARITAL_STATUS_CODE	INCOME_DESC	HOMEOWNER_DESC	HH_COMP_I
0	65+	A	35-49K	Homeowner	2 Adults No
1	45-54	A	50-74K	Homeowner	2 Adults No

In [222…

```
demographic_RAW['AGE_DESC'].value_counts()
```

Out[222…
```
AGE_DESC
45-54    901
35-44    594
25-34    445
65+      230
55-64    187
19-24    143
Name: count, dtype: int64
```

In [223…
```
demographic_RAW['HOUSEHOLD_SIZE_DESC'].value_counts()
```

Out[223…
```
HOUSEHOLD_SIZE_DESC
2     1004
1      783
3      338
5+     206
4      169
Name: count, dtype: int64
```

In [224…
```
CAMPAIGN_DESC_RAW.head(5)
```

Out[224…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration
0	TypeB	24	2020-10-21	2020-12-20	60 days
1	TypeC	15	2020-07-01	2020-12-09	161 days
2	TypeB	25	2020-10-21	2020-11-22	32 days
3	TypeC	20	2020-09-07	2020-11-16	70 days
4	TypeB	23	2020-10-08	2020-11-15	38 days

In [225…
```
CAMPAIGN_DESC_RAW.dtypes
```

Out[225…
```
DESCRIPTION                   object
CAMPAIGN                        int8
START_DAY             datetime64[ns]
END_DAY               datetime64[ns]
Campaign_Duration    timedelta64[ns]
dtype: object
```

In [226…
```
CAMPAIGN_DESC_RAW.Campaign_Duration.mean()
```

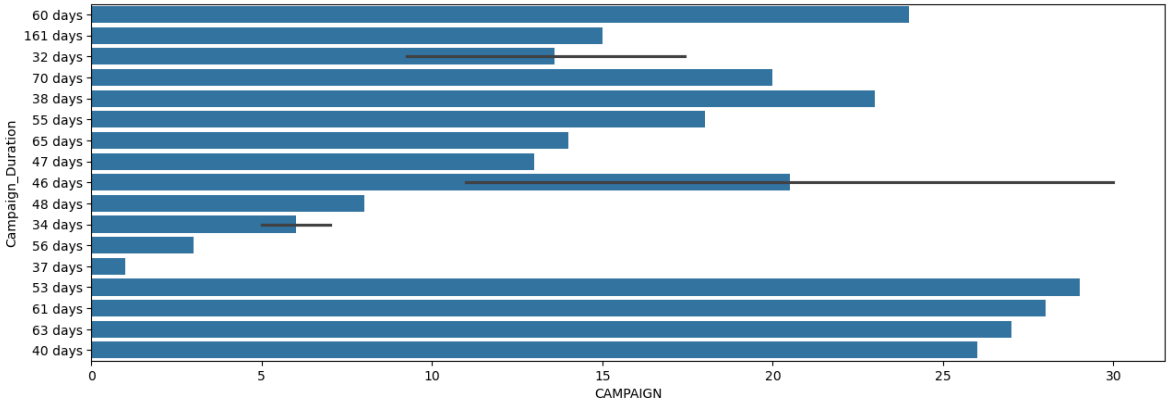
Out[226…
```
Timedelta('46 days 14:24:00')
```

In [227…
```
CAMPAIGN_DESC_RAW['Campaign_Duration'].mean()
```

Out[227…
```
Timedelta('46 days 14:24:00')
```

The Average Campaign Duration is 46.6 days

In [228…
```
%matplotlib inline
plt.figure(figsize=(15,5))
sns.barplot(x='CAMPAIGN',y='Campaign_Duration',data = CAMPAIGN_DESC_RAW)
plt.show()
```



campaign lasted more than 160 days

In [229…
```
CAMPAIGN_DESC_RAW.groupby('DESCRIPTION').aggregate({'CAMPAIGN':'count','Campaign
```

Out[229…

DESCRIPTION	CAMPAIGN	Campaign_Duration
TypeA	5	47 days 04:48:00
TypeB	19	37 days 15:09:28.421052631
TypeC	6	74 days 12:00:00

In [ ]:
```

```

There have been 19 type B campaigns, whose average length was 38 days. In comparison, there has been 6 type C campaigns of 75 days on average.

In [230…
```
CAMPAIGN_DESC_RAW.head(2)
```

Out[230…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration
0	TypeB	24	2020-10-21	2020-12-20	60 days
1	TypeC	15	2020-07-01	2020-12-09	161 days

In [231…
```
CAMPAIGN_DESC_RAW['Start_Month']=CAMPAIGN_DESC_RAW['START_DAY'].dt.strftime('%m'
```

In [232…
```
CAMPAIGN_DESC_RAW['End_Month']=CAMPAIGN_DESC_RAW['END_DAY'].dt.strftime('%m')
```

In [233…
```
CAMPAIGN_DESC_RAW['Start_Year']=CAMPAIGN_DESC_RAW['START_DAY'].dt.strftime('%Y')
```

In [234…
```
CAMPAIGN_DESC_RAW['End_Year']=CAMPAIGN_DESC_RAW['END_DAY'].dt.strftime('%Y')
```

In [235…
```
CAMPAIGN_DESC_RAW.head(10)
```

Out[235…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration	Start_Month
0	TypeB	24	2020-10-21	2020-12-20	60 days	10
1	TypeC	15	2020-07-01	2020-12-09	161 days	07
2	TypeB	25	2020-10-21	2020-11-22	32 days	10
3	TypeC	20	2020-09-07	2020-11-16	70 days	09
4	TypeB	23	2020-10-08	2020-11-15	38 days	10
5	TypeB	21	2020-09-16	2020-10-18	32 days	09
6	TypeB	22	2020-09-16	2020-10-18	32 days	09
7	TypeA	18	2020-08-10	2020-10-04	55 days	08
8	TypeB	19	2020-08-26	2020-09-27	32 days	08
9	TypeB	17	2020-07-29	2020-08-30	32 days	07

In [236…

```
CAMPAIGN_DESC_RAW.dtypes
```

Out[236…

```
DESCRIPTION                  object
CAMPAIGN                       int8
START_DAY             datetime64[ns]
END_DAY               datetime64[ns]
Campaign_Duration    timedelta64[ns]
Start_Month                   object
End_Month                     object
Start_Year                    object
End_Year                      object
dtype: object
```

In [237…

```
CAMPAIGN_RAW.shape
```

Out[237…

```
(7208, 3)
```

In [238…

```
CAMPAIGN_RAW.head(5)
```

Out[238…

	DESCRIPTION	HOUSEHOLD_KEY	CAMPAIGN
0	TypeA	17	26
1	TypeA	27	26
2	TypeA	212	26
3	TypeA	208	26
4	TypeA	192	26

In [239…

```
CAMPAIGN_RAW.columns
```

Out[239…

```
Index(['DESCRIPTION', 'HOUSEHOLD_KEY', 'CAMPAIGN'], dtype='object')
```

In [240…

```
CAMPAIGN_RAW.isnull().sum()
```

Out[240…

```
DESCRIPTION      0
HOUSEHOLD_KEY    0
CAMPAIGN         0
dtype: int64
```

In [241…

```
CAMPAIGN_RAW['HOUSEHOLD_KEY'].nunique()
```

Out[241…

```
1584
```

There are 1584 household participated to the campaign, leaving 916 household who never participated to any campaign.

In [242…

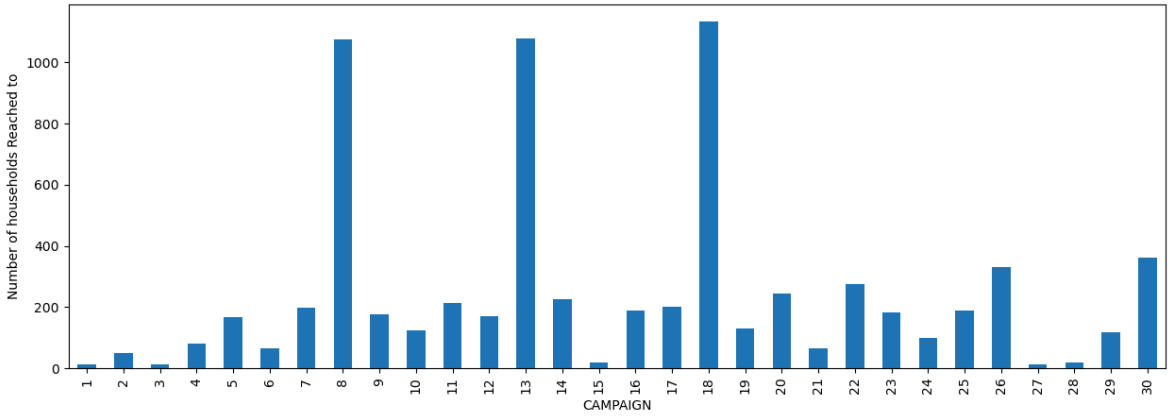
```
CAMPAIGN_RAW.groupby('HOUSEHOLD_KEY')['CAMPAIGN'].count()
```

Out[242…

```
HOUSEHOLD_KEY
1       8
2       1
3       3
4       1
6       4
       ..
2496    5
2497    3
2498    6
2499    2
2500    8
Name: CAMPAIGN, Length: 1584, dtype: int64
```

In [243…

```
%matplotlib inline
plt.figure(figsize=(15,5))
CAMPAIGN_RAW.groupby('CAMPAIGN')['HOUSEHOLD_KEY'].count().plot.bar()
plt.ylabel('Number of households Reached to')
plt.show()
```



In Compaigning 18 maximum number of household participated

In [244…
```
COUPON_RAW.shape
```

Out[244…
```
(124548, 3)
```

In [245…
```
COUPON_RAW.columns
```

Out[245…
```
Index(['COUPON_UPC', 'PRODUCT_ID', 'CAMPAIGN'], dtype='object')
```

In [246…
```
COUPON_RAW['COUPON_UPC'].nunique()
```

Out[246…
```
1135
```

In [247…
```
Coupon_Given=COUPON_RAW.groupby("CAMPAIGN").aggregate(Total_product = ('PRODUCT_
                                                      Total_Coupon_Given = ('COUPC
```

In [248…
```
Coupon_Given.sort_values(by='Total_product',ascending=False).head(10)
```

Out[248…

CAMPAIGN	Total_product	Total_Coupon_Given
13	35616	207
18	35513	209
8	17178	209
26	3396	181
30	3205	181
22	1917	21
24	1887	2
16	1025	13
25	826	17
9	825	18

In [249…
```
Coupon_Given.head(10)
```

Out[249…

CAMPAIGN	Total_product	Total_Coupon_Given
1	383	11
2	301	16
3	506	34
4	200	12
5	443	11
6	18	1
7	259	8
8	17178	209
9	825	18
10	393	14

In [250…

```
CAMPAIGN_DESC_RAW.head(2)
```

Out[250…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration	Start_Month
0	TypeB	24	2020-10-21	2020-12-20	60 days	10
1	TypeC	15	2020-07-01	2020-12-09	161 days	07

In [251…

```
Coupon_Given=Coupon_Given.merge(right=CAMPAIGN_DESC_RAW,on='CAMPAIGN',how='left'
```

In [252…

```
Coupon_Given
```

Out[252...

	CAMPAIGN	Total_product	Total_Coupon_Given	DESCRIPTION	START_DAY	END_DA
0	1	383	11	TypeB	2019-12-13	2020-0
1	2	301	16	TypeB	2019-12-18	2020-0
2	3	506	34	TypeC	2019-12-23	2020-0
3	4	200	12	TypeB	2020-01-08	2020-0
4	5	443	11	TypeB	2020-01-13	2020-0
5	6	18	1	TypeC	2020-01-29	2020-0
6	7	259	8	TypeB	2020-02-03	2020-0
7	8	17178	209	TypeA	2020-02-17	2020-0
8	9	825	18	TypeB	2020-03-11	2020-0
9	10	393	14	TypeB	2020-04-08	2020-0
10	11	278	13	TypeB	2020-04-22	2020-0
11	12	620	15	TypeB	2020-04-22	2020-0
12	13	35616	207	TypeA	2020-05-19	2020-0
13	14	714	26	TypeC	2020-06-15	2020-0
14	15	55	2	TypeC	2020-07-01	2020-1
15	16	1025	13	TypeB	2020-07-15	2020-0
16	17	713	19	TypeB	2020-07-29	2020-0
17	18	35513	209	TypeA	2020-08-10	2020-1
18	19	375	11	TypeB	2020-08-26	2020-0
19	20	609	24	TypeC	2020-09-07	2020-1
20	21	242	16	TypeB	2020-09-16	2020-1

	CAMPAIGN	Total_product	Total_Coupon_Given	DESCRIPTION	START_DAY	END_DA
21	22	1917	21	TypeB	2020-09-16	2020-1
22	23	767	18	TypeB	2020-10-08	2020-1
23	24	1887	2	TypeB	2020-10-21	2020-1
24	25	826	17	TypeB	2020-10-21	2020-1
25	26	3396	181	TypeA	2019-08-13	2019-0
26	27	632	27	TypeC	2019-08-26	2019-1
27	28	535	28	TypeB	2019-09-17	2019-1
28	29	689	33	TypeB	2019-10-09	2019-1
29	30	3205	181	TypeA	2019-11-20	2020-0

In [253…

```
Coupon_Given.columns
```

Out[253…

```
Index(['CAMPAIGN', 'Total_product', 'Total_Coupon_Given', 'DESCRIPTION',
       'START_DAY', 'END_DAY', 'Campaign_Duration', 'Start_Month', 'End_Month',
       'Start_Year', 'End_Year'],
      dtype='object')
```

In [254…

```
Coupon_Given.loc[:,('CAMPAIGN', 'Total_product', 'Total_Coupon_Given', 'DESCRIPT
       'Start_Year', 'End_Year','Campaign_Duration')].sort_values(by='Total_prod
```

Out[254…

	CAMPAIGN	Total_product	Total_Coupon_Given	DESCRIPTION	Start_Year	End_Year
12	13	35616	207	TypeA	2020	2020
17	18	35513	209	TypeA	2020	2020
7	8	17178	209	TypeA	2020	2020
25	26	3396	181	TypeA	2019	2019
29	30	3205	181	TypeA	2019	2020

In [255…

```
COUPON_RAW.head(5)
```

Out[255…

	COUPON_UPC	PRODUCT_ID	CAMPAIGN
0	10000089061	27160	4
1	10000089064	27754	9
2	10000089073	28897	12
3	51800009050	28919	28
4	52100000076	28929	25

In [256…

```
PRODUCT_RAW.head(5)
```

Out[256…

	PRODUCT_ID	MANUFACTURER	DEPARTMENT	BRAND	COMMODITY_DESC	SUB_CC
0	25671	2	GROCERY	National	FRZN ICE	ICE -
1	26081	2	MISC. TRANS.	National	NO COMMODITY DESCRIPTION	NO
2	26093	69	PASTRY	Private	BREAD	BREAD
3	26190	69	GROCERY	Private	FRUIT - SHELF STABLE	
4	26355	69	GROCERY	Private	COOKIES/CONES	SF

Campaign 13,18,8 are the one with most product in them

In [257…

```
coupon_product = COUPON_RAW.merge(right=PRODUCT_RAW,on='PRODUCT_ID',how='left')
```

In [258…

```
coupon_product
```

Out[258…

	COUPON_UPC	PRODUCT_ID	CAMPAIGN	MANUFACTURER	DEPARTMENT	BR
0	10000089061	27160	4	69	GROCERY	Pr
1	10000089064	27754	9	69	GROCERY	Pr
2	10000089073	28897	12	69	GROCERY	Pr
3	51800009050	28919	28	236	GROCERY	Nat
4	52100000076	28929	25	317	GROCERY	Nat
...	...	...	...	...	...	
124543	10000085478	13038802	18	1119	DRUG GM	Nat
124544	10000085479	13039368	18	2875	MEAT	Nat
124545	10000085478	13039553	18	1229	DRUG GM	Nat
124546	10000085479	13776332	18	4824	MEAT	Nat
124547	10000085477	13213390	18	5155	GROCERY	Nat

124548 rows × 9 columns

In [259…

```
coupon_product.isnull().sum()
```

Out[259…

```
COUPON_UPC              0
PRODUCT_ID              0
CAMPAIGN                0
MANUFACTURER            0
DEPARTMENT              0
BRAND                   0
COMMODITY_DESC          0
SUB_COMMODITY_DESC      0
CURR_SIZE_OF_PRODUCT    0
dtype: int64
```

Top 10 Product on which coupon has been applied

In [260…

```
coupon_product['COMMODITY_DESC'].value_counts().head(10)
```

Out[260…

```
COMMODITY_DESC
HAIR CARE PRODUCTS        6261
MAKEUP AND TREATMENT      5585
FRZN MEAT/MEAT DINNERS    4893
FROZEN PIZZA              3658
BEEF                      3314
ICE CREAM/MILK/SHERBTS    3257
CHEESE                    3146
YOGURT                    2460
ORAL HYGIENE PRODUCTS     2437
LUNCHMEAT                 2425
Name: count, dtype: int64
```

Most prominent products among coupons created are bathroom products such as hair care and makeup.

In [261…

```
COUPON_REDEMPT_RAW.shape
```

Out[261…

```
(2318, 5)
```

In [262…

```
COUPON_REDEMPT_RAW.columns
```

Out[262…

```
Index(['HOUSEHOLD_KEY', 'DAY', 'COUPON_UPC', 'CAMPAIGN', 'Date'], dtype='objec
t')
```

In [263…

```
COUPON_REDEMPT_RAW['COUPON_UPC'].nunique()
```

Out[263…

```
556
```

In [264…

```
COUPON_REDEMPT_RAW.head(2)
```

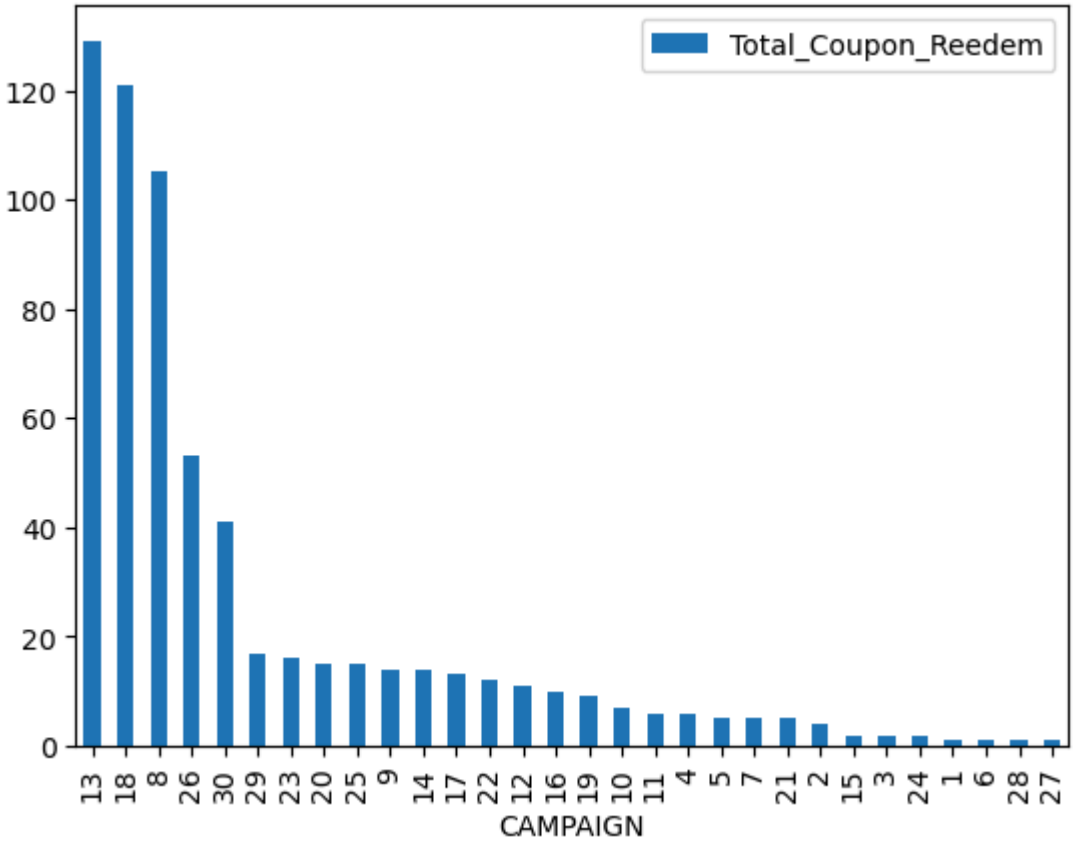
Out[264…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN	Date
0	1	421	10000085364	8	2020-02-26
1	1	421	51700010076	8	2020-02-26

Only 556 coupon has been used

In [265…

```
%matplotlib inline
COUPON_REDEMPT_RAW.groupby('CAMPAIGN').agg(Total_Coupon_Reedem=('COUPON_UPC','nu
plt.show()
```



In [266…
```
coupon_redeem = COUPON_REDEMPT_RAW.groupby('CAMPAIGN').agg(Total_Coupon_Reedem=(
```

In [267…
```
coupon_redeem.sort_values(by='Total_Coupon_Reedem',ascending=False)
```

Out[267…

CAMPAIGN	Total_Coupon_Reedem
13	129
18	121
8	105
26	53
30	41
29	17
23	16
20	15
25	15
9	14
14	14
17	13
22	12
12	11
16	10
19	9
10	7
11	6
4	6
5	5
7	5
21	5
2	4
15	2
3	2
24	2
1	1
6	1
28	1
27	1

In [268…

```
coupon_redeem.head(2)
```

Out[268…

CAMPAIGN	Total_Coupon_Reedem
1	1
2	4

In [269…

```
Coupon_Given.head(2)
```

Out[269…

	CAMPAIGN	Total_product	Total_Coupon_Given	DESCRIPTION	START_DAY	END_DAY
0	1	383	11	TypeB	2019-12-13	2020-01-19
1	2	301	16	TypeB	2019-12-18	2020-01-19

In [270…

```
coupon_redeem=coupon_redeem.merge(right=Coupon_Given,on='CAMPAIGN',how='left')
```

In [271…

```
coupon_redeem.head(2)
```

Out[271…

	CAMPAIGN	Total_Coupon_Reedem	Total_product	Total_Coupon_Given	DESCRIPTION
0	1	1	383	11	TypeB
1	2	4	301	16	TypeB

In [272…

```
coupon_redeem['Coupon_Reedemption_Rate']=(coupon_redeem['Total_Coupon_Reedem']/c
```

In [273…

```
coupon_redeem.sort_values(by='Coupon_Reedemption_Rate',ascending=False)
```

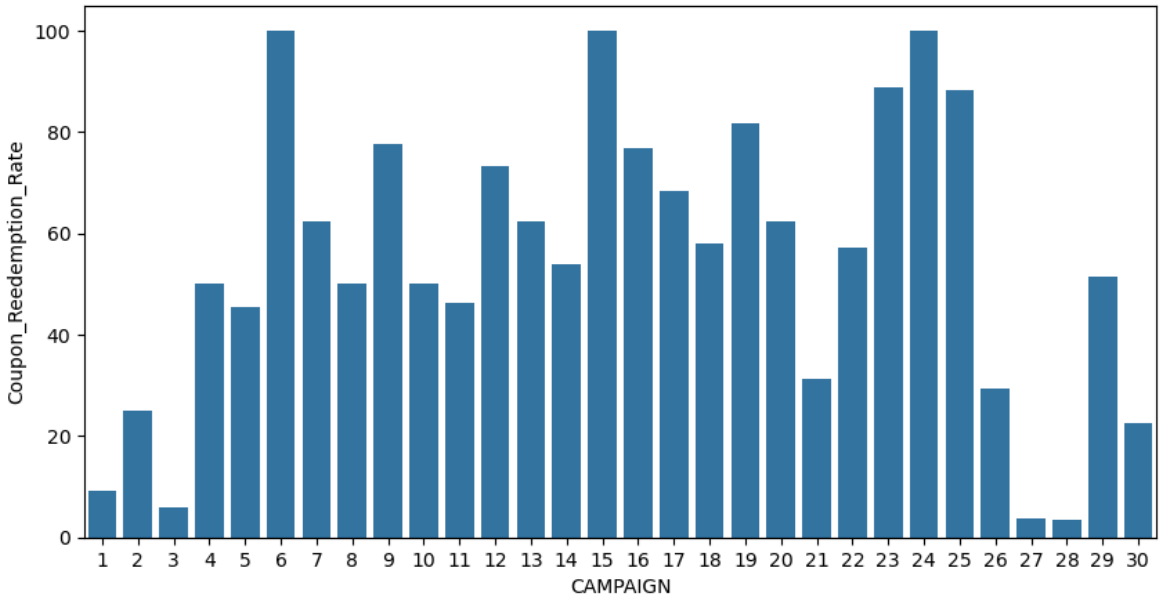
Out[273...

	CAMPAIGN	Total_Coupon_Reedem	Total_product	Total_Coupon_Given	DESCRIPTIO
5	6	1	18	1	Type
14	15	2	55	2	Type
23	24	2	1887	2	Type
22	23	16	767	18	Type
24	25	15	826	17	Type
18	19	9	375	11	Type
8	9	14	825	18	Type
15	16	10	1025	13	Type
11	12	11	620	15	Type
16	17	13	713	19	Type
19	20	15	609	24	Type
6	7	5	259	8	Type
12	13	129	35616	207	Type
17	18	121	35513	209	Type
21	22	12	1917	21	Type
13	14	14	714	26	Type
28	29	17	689	33	Type
7	8	105	17178	209	Type
3	4	6	200	12	Type
9	10	7	393	14	Type
10	11	6	278	13	Type

	CAMPAIGN	Total_Coupon_Reedem	Total_product	Total_Coupon_Given	DESCRIPTIO
4	5	5	443	11	Type
20	21	5	242	16	Type
25	26	53	3396	181	Type
1	2	4	301	16	Type
29	30	41	3205	181	Type
0	1	1	383	11	Type
2	3	2	506	34	Type
26	27	1	632	27	Type
27	28	1	535	28	Type

In [274…

```
%matplotlib inline
plt.figure(figsize=(10,5))
sns.barplot(x='CAMPAIGN',y='Coupon_Reedemption_Rate',data=coupon_redeem)
plt.show()
```



In [275…

```
TRANSACTION_RAW.head(2)
```

Out[275…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE	STOR
0	2375	26984851472	1	1004906	1	1.39	
1	2375	26984851472	1	1033142	1	0.82	

In [276…
```
TRANSACTION_RAW.shape
```

Out[276…
```
(450416, 13)
```

In [277…
```
TRANSACTION_RAW.columns
```

Out[277…
```
Index(['HOUSEHOLD_KEY', 'BASKET_ID', 'DAY', 'PRODUCT_ID', 'QUANTITY',
       'SALES_VALUE', 'STORE_ID', 'RETAIL_DISC', 'TRANS_TIME', 'WEEK_NO',
       'COUPON_DISC', 'COUPON_MATCH_DISC', 'DATE'],
      dtype='object')
```

In [278…
```
TRANSACTION_RAW.isnull().sum()
```

Out[278…
```
HOUSEHOLD_KEY        0
BASKET_ID            0
DAY                  0
PRODUCT_ID           0
QUANTITY             0
SALES_VALUE          0
STORE_ID             0
RETAIL_DISC          0
TRANS_TIME           0
WEEK_NO              0
COUPON_DISC          0
COUPON_MATCH_DISC    0
DATE                 0
dtype: int64
```

In [279…
```
TRANSACTION_RAW['BASKET_ID'].count()
```

Out[279…
```
np.int64(450416)
```

In [280…
```
TRANSACTION_RAW['BASKET_ID'].nunique()
```

Out[280…
```
50479
```

In [281…
```
TRANSACTION_RAW['HOUSEHOLD_KEY'].nunique()
```

Out[281…
```
2494
```

In [282…
```
trnx_bucket=TRANSACTION_RAW.groupby('BASKET_ID').agg({'SALES_VALUE':'sum','COUPC
```

In [283…
```
trnx_bucket.head(2)
```

Out[283…

BASKET_ID	SALES_VALUE	COUPON_DISC	COUPON_MATCH_DISC
26984851472	5.91	0	0
26984851516	12.43	0	0

In [284…

```
trnx_bucket['use_coupon']=trnx_bucket['COUPON_DISC']!=0
```

In [285…

```
trnx_bucket['use_coupon'].value_counts()
```

Out[285…

```
use_coupon
False    47892
True      2587
Name: count, dtype: int64
```

In [286…

```
trnx_bucket.sort_values('SALES_VALUE',ascending=False).head(10)
```

Out[286…

BASKET_ID	SALES_VALUE	COUPON_DISC	COUPON_MATCH_DISC	use_coupon
27534431747	465.70	-3	-1	True
27798435271	463.17	-2	-2	True
27093315740	400.79	0	0	False
28941937397	400.56	-1	0	True
28210676830	391.05	0	0	False
27950470466	376.38	-2	0	True
29034966796	369.48	0	0	False
28919755135	368.95	0	0	False
28675556859	365.86	-3	0	True
29221810662	358.98	0	0	False

In [287…

```
round(trnx_bucket['SALES_VALUE'].mean(),2)
```

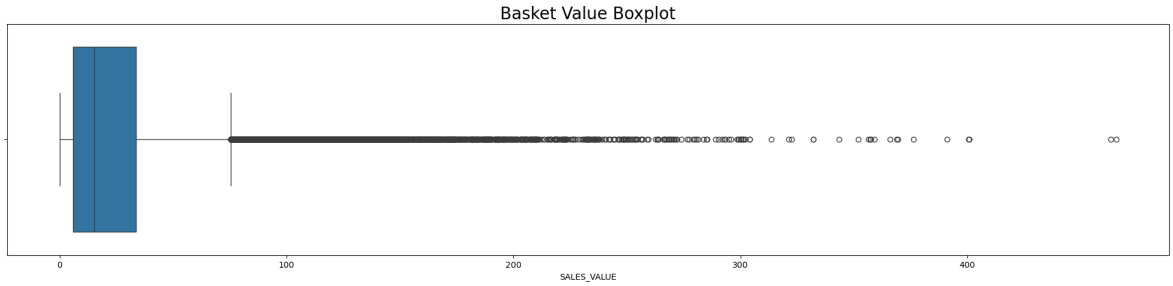
Out[287…

```
np.float64(26.93)
```

The Average Basket Value is $26.93

In [288…

```
%matplotlib inline
plt.figure(figsize=(25,5))
sns.boxplot(x=trnx_bucket['SALES_VALUE'])
plt.title('Basket Value Boxplot', fontsize=20)
plt.show()
```



In [289…
```
trnx_bucket.head(3)
```

Out[289…

BASKET_ID	SALES_VALUE	COUPON_DISC	COUPON_MATCH_DISC	use_coupon
26984851472	5.91	0	0	False
26984851516	12.43	0	0	False
26984896261	11.37	0	0	False

In [290…
```
trnx_bucket.groupby('use_coupon').agg(sales_mean=('SALES_VALUE','mean'),
                                      coupon_disc_mean=('COUPON_DISC','mean'),
                                      coupon_match_disc_mean=('COUPON_MATCH_DISC
                                      use_coupon_count=('use_coupon','count'))
```

Out[290…

use_coupon	sales_mean	coupon_disc_mean	coupon_match_disc_mean	use_coupon_coun
False	25.009980	0.000000	0.000000	4789
True	62.533348	-2.587167	-0.334364	258

The Average basket value without coupon is $26.79

The Average basket value with a coupon is $68.21

The Average discount generated by coupon is $2.98

It shows that customer purchase more product when coupon is given to them

In [291…
```
trnx_desc=TRANSACTION_RAW.merge(right=trnx_bucket,on='BASKET_ID',how='left')
```

In [292…
```
trnx_desc
```

Out[292…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE_x
0	2375	26984851472	1	1004906	1	1.39
1	2375	26984851472	1	1033142	1	0.82
2	2375	26984851472	1	1036325	1	0.99
3	2375	26984851472	1	1082185	1	1.21
4	2375	26984851472	1	8160430	1	1.50
...	...	...	...	...	...	.
450411	916	29339646926	176	887342	1	1.50
450412	916	29339646926	176	942778	1	2.99
450413	2086	29339647754	176	868764	3	4.00
450414	2086	29339647754	176	1093413	1	0.76
450415	2086	29339647754	176	1126392	1	2.83

450416 rows × 17 columns

In [293…

```
trnx_desc=trnx_desc.merge(right=PRODUCT_RAW,on='PRODUCT_ID',how='left')
```

In [294…

```
trnx_desc
```

Out[294…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE_x
0	2375	26984851472	1	1004906	1	1.39
1	2375	26984851472	1	1033142	1	0.82
2	2375	26984851472	1	1036325	1	0.99
3	2375	26984851472	1	1082185	1	1.21
4	2375	26984851472	1	8160430	1	1.50
...	...	...	...	...	...	.
450411	916	29339646926	176	887342	1	1.50
450412	916	29339646926	176	942778	1	2.99
450413	2086	29339647754	176	868764	3	4.00
450414	2086	29339647754	176	1093413	1	0.76
450415	2086	29339647754	176	1126392	1	2.83

450416 rows × 23 columns

In [295…

```
trnx_desc.drop(['SALES_VALUE_y','COUPON_DISC_y','COUPON_MATCH_DISC_y'],axis=1,in
```

In [296…

```
trnx_desc
```

Out[296…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE_x
0	2375	26984851472	1	1004906	1	1.39
1	2375	26984851472	1	1033142	1	0.82
2	2375	26984851472	1	1036325	1	0.99
3	2375	26984851472	1	1082185	1	1.21
4	2375	26984851472	1	8160430	1	1.50
...	...	...	...	...	...	.
450411	916	29339646926	176	887342	1	1.50
450412	916	29339646926	176	942778	1	2.99
450413	2086	29339647754	176	868764	3	4.00
450414	2086	29339647754	176	1093413	1	0.76
450415	2086	29339647754	176	1126392	1	2.83

450416 rows × 20 columns

In [297…

```
commodity_coupon=trnx_desc.groupby('COMMODITY_DESC').agg(Total_quantity=('QUANTI
                                                         use_coupon=('use_coupon'
                                                         coupon=('COUPON_DISC_x',
```

In [298…

```
commodity_coupon.sort_values(by='use_coupon',ascending=False).head(10)
```

Out[298…

COMMODITY_DESC	Total_quantity	use_coupon	coupon
SOFT DRINKS	22020	2623	-1324
BAKED BREAD/BUNS/ROLLS	14999	1687	-66
CHEESE	12202	1642	-54
FLUID MILK PRODUCTS	14462	1455	-51
BAG SNACKS	11974	1349	-17
FRZN MEAT/MEAT DINNERS	9340	1343	-77
YOGURT	7897	1174	-104
FROZEN PIZZA	7719	1167	-117
BABY FOODS	3664	1114	-38
BEEF	8680	980	0

In [299…

```
commodity_coupon['coupon%']=round((commodity_coupon['use_coupon']/commodity_coup
```

In [300…

```
commodity_coupon
```

Out[300…

COMMODITY_DESC	Total_quantity	use_coupon	coupon	coupon%
	1303	872	-1396	66.92
(CORP USE ONLY)	12	0	0	0.00
ADULT INCONTINENCE	74	3	-1	4.05
AIR CARE	928	193	-124	20.80
ANALGESICS	681	89	-3	13.07
...	...	...	...	...
WAREHOUSE SNACKS	1846	213	-18	11.54
WATCHES/CALCULATORS/LOBBY	12	0	0	0.00
WATER	363	36	0	9.92
WATER - CARBONATED/FLVRD DRINK	4188	371	-13	8.86
YOGURT	7897	1174	-104	14.87

298 rows × 4 columns

In [301…

```
commodity_coupon.sort_values(by='coupon%',ascending=False).head(10)
```

Out[301…

COMMODITY_DESC	Total_quantity	use_coupon	coupon	coupon%
NATURAL VITAMINS	1	1	-5	100.00
	1303	872	-1396	66.92
VALENTINE	5	2	0	40.00
BABY FOODS	3664	1114	-38	30.40
RW FRESH PROCESSED MEAT	7	2	0	28.57
DIAPERS & DISPOSABLES	864	231	-215	26.74
BABY HBC	817	181	-82	22.15
AIR CARE	928	193	-124	20.80
FRZN BREAKFAST FOODS	1743	347	-61	19.91
CAT LITTER	531	104	-11	19.59

The Most prominents product for which coupon were created are haircare and makeup products,coupons mostly used in food items

In [302…

```
TRANSACTION_RAW.head(2)
```

Out[302…

	HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY	SALES_VALUE	STOR
0	2375	26984851472	1	1004906	1	1.39	
1	2375	26984851472	1	1033142	1	0.82	

In [303…

```
TRANSACTION_RAW.groupby(['HOUSEHOLD_KEY','WEEK_NO','DAY']).agg({'SALES_VALUE':'s
```

Out[303…

HOUSEHOLD_KEY	WEEK_NO	DAY	SALES_VALUE	RETAIL_DISC	COUPON_DISC	COUPOI
1	**8**	**51**	78.66	-16.54	-1	
	10	**67**	41.10	-8.59	0	
	13	**88**	26.90	-6.72	0	
	14	**94**	63.43	-11.08	-2	
	15	**101**	53.45	-16.42	0	
...	**...**	**...**	...	...	...	
2500	**21**	**145**	50.23	-2.02	-3	
	22	**150**	55.03	-2.17	0	
	26	**174**	110.30	-49.05	-1	
		175	10.59	0.00	-2	
		176	20.26	-2.51	0	

40506 rows × 4 columns

In [304…

```
TRANSACTION_RAW.groupby(TRANSACTION_RAW['DATE'].dt.year).agg({'SALES_VALUE':'sum
```

Out[304…

DATE	SALES_VALUE	RETAIL_DISC	COUPON_DISC
2019	1359551.73	-248978.11	-6693

Sales value increases as Retail discout and coupon discount increases

Dropping the Columns

In [305…

```
from datetime import datetime
```

In [306…

```
CAMPAIGN_DESC_RAW.head(2)
```

Out[306…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration	Start_Month
0	TypeB	24	2020-10-21	2020-12-20	60 days	10
1	TypeC	15	2020-07-01	2020-12-09	161 days	07

In [308…

```
type(CAMPAIGN_DESC_RAW['START_DAY'])
```

Out[308…

```
pandas.core.series.Series
```

In [310…
```
CAMPAIGN_DESC_RAW.dtypes
```

Out[310…
```
DESCRIPTION                   object
CAMPAIGN                        int8
START_DAY             datetime64[ns]
END_DAY               datetime64[ns]
Campaign_Duration    timedelta64[ns]
Start_Month                   object
End_Month                     object
Start_Year                    object
End_Year                      object
dtype: object
```

In [311…
```
CAMPAIGN_DESC_RAW['START_DAY']=pd.to_datetime(CAMPAIGN_DESC_RAW['START_DAY']).ap
```

In [312…
```
type(CAMPAIGN_DESC_RAW['START_DAY'])
```

Out[312…
```
pandas.core.series.Series
```

In [313…
```
CAMPAIGN_DESC_RAW['START_DAY']
```

Out[313…
```
0     2020-10-21
1     2020-07-01
2     2020-10-21
3     2020-09-07
4     2020-10-08
5     2020-09-16
6     2020-09-16
7     2020-08-10
8     2020-08-26
9     2020-07-29
10    2020-06-15
11    2020-07-15
12    2020-05-19
13    2020-04-22
14    2020-04-22
15    2020-04-08
16    2020-03-11
17    2020-02-17
18    2020-02-03
19    2020-01-29
20    2019-12-23
21    2020-01-13
22    2020-01-08
23    2019-12-13
24    2019-12-18
25    2019-11-20
26    2019-10-09
27    2019-09-17
28    2019-08-26
29    2019-08-13
Name: START_DAY, dtype: object
```

In [314…
```
CAMPAIGN_DESC_RAW.dtypes
```

Out[314…
```
DESCRIPTION                   object
CAMPAIGN                        int8
START_DAY                     object
END_DAY                datetime64[ns]
Campaign_Duration     timedelta64[ns]
Start_Month                   object
End_Month                     object
Start_Year                    object
End_Year                      object
dtype: object
```

In [315…
```
CAMPAIGN_DESC_RAW['END_DAY']=pd.to_datetime(CAMPAIGN_DESC_RAW['END_DAY']).apply(
```

In [316…
```
CAMPAIGN_DESC_RAW.dtypes
```

Out[316…
```
DESCRIPTION                   object
CAMPAIGN                        int8
START_DAY                     object
END_DAY                       object
Campaign_Duration     timedelta64[ns]
Start_Month                   object
End_Month                     object
Start_Year                    object
End_Year                      object
dtype: object
```

In [317…
```
COUPON_REDEMPT_RAW.head(2)
```

Out[317…

	HOUSEHOLD_KEY	DAY	COUPON_UPC	CAMPAIGN	Date
0	1	421	10000085364	8	2020-02-26
1	1	421	51700010076	8	2020-02-26

In [318…
```
COUPON_REDEMPT_RAW.drop(['DAY'],axis=1,inplace=True)
```

In [319…
```
COUPON_REDEMPT_RAW.head(2)
```

Out[319…

	HOUSEHOLD_KEY	COUPON_UPC	CAMPAIGN	Date
0	1	10000085364	8	2020-02-26
1	1	51700010076	8	2020-02-26

In [320…
```
COUPON_REDEMPT_RAW.dtypes
```

Out[320…
```
HOUSEHOLD_KEY             int16
COUPON_UPC                int64
CAMPAIGN                   int8
Date             datetime64[ns]
dtype: object
```

In [321…
```
COUPON_REDEMPT_RAW['Date']=pd.to_datetime(COUPON_REDEMPT_RAW['Date']).apply(lamb
```

In [322…
```
COUPON_REDEMPT_RAW.dtypes
```

Out[322…

```
HOUSEHOLD_KEY     int16
COUPON_UPC        int64
CAMPAIGN           int8
Date             object
dtype: object
```

In [325…

```
TRANSACTION_RAW.drop(['DAY','WEEK_NO'],axis=1,inplace=True)
```

In [326…

```
TRANSACTION_RAW.head()
```

Out[326…

	HOUSEHOLD_KEY	BASKET_ID	PRODUCT_ID	QUANTITY	SALES_VALUE	STORE_ID
0	2375	26984851472	1004906	1	1.39	364
1	2375	26984851472	1033142	1	0.82	364
2	2375	26984851472	1036325	1	0.99	364
3	2375	26984851472	1082185	1	1.21	364
4	2375	26984851472	8160430	1	1.50	364

In [327…

```
TRANSACTION_RAW.dtypes
```

Out[327…

```
HOUSEHOLD_KEY                int16
BASKET_ID                    int64
PRODUCT_ID                   int32
QUANTITY                     int32
SALES_VALUE                float64
STORE_ID                     int16
RETAIL_DISC                float64
TRANS_TIME                   int16
COUPON_DISC                   int8
COUPON_MATCH_DISC             int8
DATE                datetime64[ns]
dtype: object
```

In [332…

```
TRANSACTION_RAW['DATE']=pd.to_datetime(TRANSACTION_RAW['DATE']).apply(lambda x:
```

In [333…

```
TRANSACTION_RAW.dtypes
```

Out[333…

```
HOUSEHOLD_KEY          int16
BASKET_ID              int64
PRODUCT_ID             int32
QUANTITY               int32
SALES_VALUE          float64
STORE_ID               int16
RETAIL_DISC          float64
TRANS_TIME             int16
COUPON_DISC             int8
COUPON_MATCH_DISC       int8
DATE                  object
dtype: object
```

Now Loading the table to DB

In [337…
```
from sqlalchemy import create_engine
from sqlalchemy.engine import url
import snowflake.connector as snowCtx
from snowflake.connector.pandas_tools import write_pandas
import pandas as pd
import getpass
```

In [347…
```
conn = snowflake.connector.connect(
        user = 'Moni',
        ##password = getpass.getpass('Your Snowflake Password: ')
        password='Moni@1234567890',
        account = 'CNXLYPS-NB32912',
        database='RETAILS',
        warehouse='COMPUTE_WH',
        SCHEMA='PUBLIC',
        schema ='PUBLIC'
 )
```

In [348…
```
cur=conn.cursor()
```

In [349…
```
CAMPAIGN_DESC_RAW.head(2)        # clean raw dataframe
```

Out[349…

	DESCRIPTION	CAMPAIGN	START_DAY	END_DAY	Campaign_Duration	Start_Month
0	TypeB	24	2020-10-21	2020-12-20	60 days	10
1	TypeC	15	2020-07-01	2020-12-09	161 days	07

In [351…
```
success, nchunks, nrows, _ = write_pandas(conn, CAMPAIGN_DESC_RAW,'CAMPAIGN_DESC
print(str(success)+','+str(nchunks)+','+str(nrows))
```

```
True,1,30
```

In [352…
```
success, nchunks, nrows, _ = write_pandas(conn, COUPON_REDEMPT_RAW,'COUPON_REDEM
print(str(success)+','+str(nchunks)+','+str(nrows))
```

```
True,1,2318
```

In [353…
```
success, nchunks, nrows, _ = write_pandas(conn,TRANSACTION_RAW ,'TRANSACTION_NEW
print(str(success)+','+str(nchunks)+','+str(nrows))
```

```
True,1,450416
```

In [354…
```
cur.close()
conn.close()
```

In [ ]: